



REQUEST FOR QUOTATION



Client: Electric Hydrogen	Rev 0	Date 05/25/23	Issued For Bid	By AR	Chk'd MHC	App'd RP	Originated By: A. Rajtar	Date: 05/16/23
Project No.: 22MH194							Document No.: 22MH194-PR8027	
Project: Product V1 Hydrogen Plant							RFQ Due Date: June 8, 2023 @ Noon (CT)	

**ELECTRIC HYDROGEN
PRODUCT V1 HYDROGEN PLANT
REQUEST FOR QUOTATION
WATER PURIFICATION SYSTEM
MAY 25, 2023**

TABLE OF CONTENTS

1. RFQ OVERVIEW	3
2. GLOSSARY OF TERMS AND DEFINITIONS	4
3. INSTRUCTIONS TO BIDDERS	6
4. TECHNICAL INSTRUCTIONS	9
5. BID ACKNOWLEDGEMENT	10
6. QUOTATION SUMMARY	11
7. SUPPLIER DOCUMENT REQUIREMENTS LIST	14
8. INSPECTION MATRIX	17
9. SPECIFICATION	19

1. RFQ OVERVIEW

M&H Consulting (MHC) is helping our customer, Electric Hydrogen (EH2), to design, procure and construct a hydrogen electrolysis facility. This RFQ is in support of that effort. We are asking your company to provide a quotation to supply Water Purification System for this facility in accordance with the following documentation.

Customer Name:	EH2
Project Name:	V1 Hydrogen Plant
Subject RFQ:	Water Purification System
Bid Acknowledgment Date:	May 30, 2023
Clarification Due Date:	June 1, 2023
Bid Due Date:	June 8, 2023 @ Noon (CT)
Address Proposal to:	Electric Hydrogen Co. 1 Strathmore Road Natick, MA 01760
Material Destination:	TBD
Target Delivery Date:	September 1, 2023
Bid Validity Requested:	30 Days
Procurement Contact:	22mh194procurement@mhes.com
Package Engineer:	Mark.Heston-Cook@mhes.com
Company Contact:	joe@eh2.com

2. GLOSSARY OF TERMS AND DEFINITIONS

2.1. Overview

This document contains definitions of common terms used throughout M&H Consulting's (MHC) standard documents and specifications.

2.2. Definitions / Abbreviations

- 2.2.1. COMPANY shall be defined as Electric Hydrogen (EH2).
- 2.2.2. CERTIFYING AUTHORITY (CA) shall be defined as all governmental agencies, quasi-government agencies, departments, or subdivisions and all federal, state, national, regional, municipal, local, or ecclesiastical authorities with jurisdiction over the WORK (i.e., American Bureau of Shipping (ABS) / DNV / Lloyds).
- 2.2.3. PURCHASER shall be defined as EH2.
- 2.2.4. ENGINEERING CONSULTANT shall be defined as M&H Consulting (MHC).
- 2.2.5. BIDDER shall mean a company that has been solicited to make a proposal to do work for, or supply equipment or materials to COMPANY.
- 2.2.6. SUPPLIER or CONTRACTOR shall be defined as the party holding the Purchase Order from the PURCHASER to supply goods, fabrication, or services for the COMPANY.
- 2.2.7. SUB-SUPPLIER or SUB-CONTRACTOR shall be defined as the party holding a Purchase Order from the SUPPLIER or CONTRACTOR.
- 2.2.8. TRANSPORTATION CONTRACTOR shall be defined as the party awarded a contract to transport equipment / materials/ structures to a defined destination.
- 2.2.9. INSPECTOR or AUTHORIZED INSPECTOR (AI) shall be a qualified individual appointed by the PURCHASER to perform inspections.
- 2.2.10. LISTED shall refer to equipment appearing on a list maintained by a nationally recognized testing laboratory (NTRL) such as UL, FM, or other testing laboratory approved by ENGINEERING CONSULTANT.
- 2.2.11. FREE ISSUED ITEMS shall mean the materials, tools, and equipment to be provided to CONTRACTOR/ SUPPLIER to complete the work.
- 2.2.12. WORK shall mean equipment, materials, fabrication, parts, software, documents, manuals, and engineering and design as required to complete the Scope of Supply.
- 2.2.13. APPROVAL or APPROVED shall mean approval or approved by an authorized representative of COMPANY or ENGINEERING CONSULTANT in writing.

2.2.14. SPECIFICATION shall mean the specification referenced setting forth the detailed requirements of the WORK.

2.2.15. THIRD PARTY shall mean any person not included in COMPANY, ENGINEERING CONSULTANT, or CONTRACTOR groups.

3. INSTRUCTIONS TO BIDDERS

1. The bid will be accepted by **email**, submitted to the following:
 - Procurement – 22mh194procurement@mhes.com
 - Joe Hodge – Joe@eh2.com
2. Within **two business days** of receipt of this inquiry, BIDDER shall complete the Inquiry Acknowledgment (see Section 5 – Bid Acknowledgment) and return to 22mh194procurement@mhes.com.
3. All clarification requests must be submitted by the clarification date and shall be directed in writing to:
 - Procurement – 22mh194procurement@mhes.com
 - Mark Heston-Cook – Mark.Heston-Cook@mhes.com
4. Bids must be received prior to the bid due date and time to be considered.
5. **Verbal clarifications will not be accepted.** Any third-party sub-contractor quoting any portion of the work to the BIDDER should submit their clarifications through the BIDDER and not directly to MHC or the COMPANY. Failure to comply may result in disqualification of BIDDER quotation.
6. BIDDERS proposal shall reflect delivery by the Target Delivery date. If not possible, BIDDER to offer their best possible delivery and any suggestions that will optimize design and reduce lead times.
7. BIDDER shall review the provided COMPANY Terms & Conditions. BIDDER shall accept the terms in full or provided redline, native comments in “Track Changes” mode.
8. COMPANY reserves the right to accept any bid, to reject any or all bids, or to negotiate contract terms with any individual BIDDER when such is deemed by COMPANY to be in its best interest. COMPANY also reserves the sole right to award to more than one BIDDER or award in full, or in part, any of the requirements.
9. Award of the contract will be made to the responsible BIDDER, whose bid conforms to the solicitation and is the most advantageous to COMPANY, considering the technical, management, schedule, and price related factors.
10. Instructions to BIDDER(S) and all other documents, datasheets, specifications, drawings, etc., noted in and attached to this RFQ are hereby made an integral part of the RFQ documents.
11. All drawings, specifications, and technical information provided to BIDDER by COMPANY shall be treated as confidential and remain property of COMPANY. BIDDER agrees not to permit any such drawing, specifications, or information to be shown or disclosed to anyone other than those who need it in connection with the work involved, or to use for any purpose other than performance of this bid.
12. In the event of conflict between the datasheets and specifications, BIDDER is to notify COMPANY in writing for clarification. If conflicts exist between the datasheets /

specifications and local, state, and / or federal codes, the more stringent of the two shall govern.

13. Pricing shall be firm through delivery and shall be inclusive of all labor, equipment, materials, consumables, and services required by the BIDDER and his SUB-CONTRACTOR(S) to furnish a completed package as outlined in the RFQ documents.
14. During the course of fabrication, the COMPANY reserves the right to make minor changes in fabrication details. Such changes will be made by the CONTRACTOR without invalidating the agreement and without additional cost to the COMPANY, provided that notice of change is given before the particular part of the work has begun. All extra work requested by the COMPANY will be performed by the CONTRACTOR on either a "Time and Material" basis, or on a "Lump Sum" basis for a price agreed upon in writing prior to the work being performed.
15. Submission of a quotation constitutes that the BIDDER has read and is completely familiar with all facets of the work as described in the RFQ documents. Any unfamiliarity with the RFQ documents shall not be considered as a basis for future price adjustments.
16. BIDDER shall perform all work and work products described in the RFQ documents. No part or portion of the work is to be subcontracted without prior written approval. If subcontracting is contemplated, BIDDER must note its intention to do so in the quotation. Breach of this clause may result in termination of the contract.
17. CONTRACTOR shall provide COMPANY or its designated representative access to any and all parts of the fabrication yard / facilities engaged in the execution of the Work.
18. Any taxes, licenses, fees, assessments, permits, or other charges incurred by the BIDDER shall be included in the quoted price. Neither the COMPANY nor its representative(s) will be responsible for reimbursing any additional amounts requested by the BIDDER for such items.
19. All guarantees and / or warranties and all rights must be fully transferable to COMPANY with the warranty period requested in the RFQ documents.
20. BIDDER shall submit, in detail, BIDDER'S technical performance guarantee (where applicable). The performance guarantee will be negotiated and agreed upon by all parties prior to award of the contract / purchase order and will be clearly stated on the Purchase Order.
21. BIDDER'S quote shall include preparation and submittal of bi-weekly progress reports, indicating status of engineering, procurement, fabrication, and scheduled ship / completion dates.
22. A detailed work schedule will be required within two weeks after receipt of order. Thereafter, the schedule shall be milestone and accompany the bi-weekly report. This schedule will detail all major milestones and events, including inspections and tests, throughout the entire period associated with the order. A two-week period shall be built into this schedule for COMPANY / COMPANY'S Representative to review drawings.

-
23. During the course of project execution, CONTRACTOR will be required to alert COMPANY of any external factors resulting in a delay in CONTRACTOR'S planned schedule.
24. The BIDDER shall submit with the quotation a bar chart production schedule showing the anticipated project duration with the minimum milestone dates as listed below:
- Receipt of Major Materials
 - Start of Fabrication
 - Completion of Fabrication
 - Completion of Testing, and Assembly
 - Shipping Date
25. Cancellation charges shall be provided in accordance with the following schedule:
- Engineering Completed
 - Materials Ordered
 - Material Received
 - Fabrication Started
26. If BIDDER has multiple fabrication facilities, BIDDER is to offer the facility nearest to the end destination to minimize shipping costs. Should facility selection impact the order's lead time, BIDDER should submit both options with accompanying shipping cost and lead time impacts for review to COMPANY.
27. CONTRACTOR shall be responsible for providing any bracing, clips, and / or brackets necessary for temporary support during transportation. CONTRACTOR shall consider the forces imposed during transit or handling, and climatic conditions in the design, construction, materials, and workmanship of all packaging. CONTRACTOR shall also provide the magnitude of all temporary loads at the support points. Final details of temporary bracing, clips, and brackets shall be subject to approval by COMPANY.
28. All piping materials (pipe and fittings) shall be supplied as new and with certified material test reports and certificates.
29. The CONTRACTOR shall furnish COMPANY Inspector with one copy of all mill certificates for all CONTRACTOR purchased materials.
30. Note that 10% of the contract / P.O. value will be held until approval of the Final CONTRACTOR Data as noted in the CONTRACTOR Data Requirements List.
31. Marking or tagging instructions shown on specifications and datasheets must be complied with in every detail.
32. BIDDER shall provide with quote field service rates and an estimate of all start-up and commissioning support required, if applicable.
33. BIDDER shall provide itemized pricing for any special tools required during installation, pre-commissioning, commissioning, and operation.

4. TECHNICAL INSTRUCTIONS

1. Country of origin of all pipe, valves, fittings and plate materials shall be US, Japan, Canada, or Western European. Valve bodies cast from India or China will be permitted. Any conflicts must be submitted in writing and require COMPANY approval in writing.

5. BID ACKNOWLEDGEMENT

Within two (business) days of receipt, complete and return this form by email to:

RFQ Communication at 22mh194procurement@mhcs.com.

INTENT TO BID FOR THE WATER PURIFICATION SYSTEM AS DESCRIBED IN RFQ NO. 22MH194-PR8027:

() We will submit our bid to you on or before June 8, 2023 @ Noon (CT).

() We will not submit a bid; the Inquiry is returned herewith.

Reason for declining to bid:

The following documents are requested:

Company:

Company Name:

Title:

E-mail:

Phone:

Address:

Signed:

Date:

6. QUOTATION SUMMARY

Please ensure a completed, native version of this document is returned with your proposal.



QUOTATION SUMMARY
WATER PURIFICATION SYSTEM
22MH194-PR8027



This form must be completed and returned in native form as part of the submitted BIDDER's quotation package in order to be considered.

BIDDER INFORMATION		
Company Name:		
Technical Representative:		
Title:		
Phone Number:		
E-mail Address:		
Quotation Number:		
Date:		
EQUIPMENT	Unit Cost	Total Cost
Engineering, Design, Drawings, Calculations, Fabrication, and Testing of WATER PURIFICATION SYSTEM per the provided specification in the RFQ package.	\$0.00	\$0.00
TOTAL COST		\$0.00
DELIVERIES		
Time from PO Issuance to Submittal of Drawings for Approval:		WEEKS
Time Allotted for Customer Review:		WEEKS
Time from Receipt of Approved for Fabrication Drawings to Readiness to Ship:		WEEKS
Time from Order Delivery to Submittal of For Record Contractor Documentation:		WEEKS
SHIPPING INFORMATION		
Shipping Point / Point of Manufacture:		
Total Estimated Shipping Weight:		
Estimated Shipping Dimensions:		
MISCELLANEOUS		
Validity Period of Quotation:		DAYS
Warranty Period Offered:		MONTHS
Requested Milestone Payments:		
Cancellation Charges:		

CHECKLIST (Highlight Yes/No Cells for Answer)		
Does BIDDER agree to provide all 2D or 3D (or other design software) files during development of the equipment package and "As-Built" at the end of fabrication?	YES	NO
Is quotation in accordance with specifications, drawings, and datasheets set forth in the Inquiry? <i>If NO, all exceptions are to be listed as required by the Instructions to Bidders.</i>	YES	NO
Is Quotation in accordance with Instructions to Bidders as set forth in the Inquiry? <i>If NO, all exceptions are to be listed as required by the Instructions to Bidders.</i>	YES	NO
Does BIDDER intend to subcontract any part of the work? <i>If YES, indicate the name(s) of the prospective third-party company(ies) and the service(s) which will be provided.</i>	YES	NO

7. SUPPLIER DOCUMENT REQUIREMENTS LIST

- BIDDERS quotation shall include time and cost to provide vendor documentation. The approved SUPPLIER will be required to provide documentation according to the following details:
 - All documentation must be marked with complete Purchase Order Number and Item / Tag number.
 - SUPPLIER'S vendor documentation will be emailed, as scheduled, to:
 - Electric Hydrogen
 - c/o M&H Consulting
 - Reference: Purchase Order Number and SDRL Description
 - Email: vendordata@mhes.com.

SUPPLIER will be held liable for all costs resulting from errors in the SUPPLIER's certified correct prints. The review and approval of SUPPLIER provided documents is for ascertaining that the SUPPLIER is fulfilling its committed obligations as perceived by M&H in accordance with the project documents. The approval of SUPPLIER provided Drawings or receipt of SUPPLIER design calculations does not constitute acceptance or verification by that the equipment satisfies all the codes, standards, technical specifications, and safety requirements as detailed in the project documents. The responsibility for technical and legal compliance for all referenced codes, technical specifications and safety standards will be the sole responsibility of the SUPPLIER. Any comments by M&H that inadvertently or intentionally violate any codes or safety standards will be brought to the attention of M&H in writing and corrected by the SUPPLIER.

7.1. Approval Documentation

Documents will be returned marked as follows:

1. Approved for Fabrication without Comments
2. Approved for Fabrication with Comments; Revise and Resubmit
3. Not Approved for Fabrication; Revise and Resubmit
4. For Information Only
5. Certified Final Vendor Data

7.2. Final Certified Documentation

- One electronic copy of all final documentation to be assembled and submitted as categorized in document request.
- Certified "As-Built" Drawings and data shall be the final versions reflecting the As-Built condition.
- **MHC will not consider the P.O. to be complete or final payment be made by the COMPANY until all document requirements have been met and approved.**

Category and Description	Required with Proposal	Date Approval Documents Required After Receipt of Order	Date Final Certified Documents Required After Shipment
Standard Information			
Drawing / Document Index		4 weeks	2 weeks
Milestone / Fabrication Schedule	Yes	Updated bi-weekly	
Quality Plan	Yes		
Approval / Comments on Provided Package Inspection Test Plan	Yes		
Shipping Plans and / or Procedures		12 weeks	
Welder Procedures / Specifications / Qualifications		4 weeks	
Stainless Steel fabrication / handling procedures		4 weeks	
Calculations and Datasheets			
Equipment Sizing Calculations			
ASME Code Calculations			
Structural Calculations			
Lifting Lug Calculations / Lifting Analysis (As Required)			
Vibration / Acoustic Analysis			
Completed Equipment Datasheets			
Instrument Datasheets			
Drawings / Bill of Materials			
AutoCAD / 2D or 3D Model Files		As Developed & Updated	Final Version Submitted with Final Vendor Data
Process Flow Diagrams	Yes – basic PFD of system	3 weeks	2 weeks
Piping & Instrumentation Diagrams		6 weeks	2 weeks
General Arrangement		8 weeks	2 weeks
Assembly Plan and Elevations		8 weeks	2 weeks
Structural Fabrication Drawings (Skid, Platforms, etc.)			
Bill of Materials			
Auxiliary Equipment Drawings / Data		8 weeks	2 weeks
Instrumentation Details		8 weeks	2 weeks

Category and Description	Required with Proposal	Date Approval Documents Required After Receipt of Order	Date Final Certified Documents Required After Shipment
Schematics / Wiring Drawings		8 weeks	2 weeks
Control Panel Drawings		8 weeks	2 weeks
Manufactured Component Data (Descriptive Literature, Operating Instructions, and Details)			
Instrumentation		10 weeks	2 weeks
Piping Specialty Items		10 weeks	2 weeks
Valves Actuated		10 weeks	2 weeks
Valves Manual		10 weeks	2 weeks
Motor Data with HP & Loads		10 weeks	2 weeks
Certificates, Reports, and Records			
Material Test Reports / Traceability (As Required)			2 weeks
Paint / Coating Reports			2 weeks
Hydrostatic Test Reports			2 weeks
NDE – Includes RT, UT, MT, & Associated Weld Maps			2 weeks
Performance Curves or Spreadsheets			2 weeks
Manufacturers Test Data			2 weeks
Performance / Acceptance Tests (SAT / FAT)			2 weeks
UL / NRTL Certificates for All Electrical			2 weeks
Weight / COG Report		12 weeks	2 weeks
Installation, Operating, and Maintenance Procedures			
Preservation / Preventive Maintenance Procedures			2 weeks
Pre-Commissioning Check Lists			2 weeks
Start-up and Commissioning Procedures			2 weeks
Installation, Operating and Maintenance Procedures			2 weeks
Recommended Start-up Spare Parts (Priced)	Yes		
Recommended Spare Parts for 2 Year Operation (Priced)	Yes		

8. INSPECTION MATRIX

As noted in the Instructions to Bidders, CONTRACTOR shall provide COMPANY or its designated representative access to any and all parts of the fabrication yard / facilities engaged in the execution of the Work. COMPANY will conduct periodic inspections on equipment as detailed in the Inspection Matrix as detailed in the following table. **CONTRACTOR shall include with their proposal any costs associated with these inspection visits. Costs for inspection efforts after award will not be accepted by COMPANY.**



INSPECTION MATRIX



Client: Electric Hydrogen	Rev. 0	Date 5/24/2023	Issued For Bid	By AR	Chk'd MHC	App'd RP	Originated By: A. Rajtar	Date: 05/23/23
Project No.: 22MH194							Document No.: 22MH194-PR8027-IM-EH2	
Project: Product V1 Hydrogen Plant							Package Name: Water Purification System	

This matrix shows the general extent of inspection to be performed by COMPANY on this work. **COMPANY reserves the right to increase or reduce these requirements at any time with or without notice.** COMPANY inspection shall be for the purpose of ensuring the specification requirements are being met and shall not be considered a substitute for CONTRACTOR's normal QA/QC activities.

The Inspection Columns are as follows:

- 1 - Witness (W) or verify (V) by COMPANY Inspector.
- 2 - Review (R) of documents by COMPANY Inspector.
- 3 - Documents (D) provided by CONTRACTOR.
- 4 - Hold Points (H) where fabrication MUST be inspected before work continues.

No.	Activities	1	2	3	4	Comments
1	Material Test Reports			D		
2	Welding Procedures		R			
3	Welder Qualification Records		R			
4	Workmanship, Material, Components, Dimensions	V				
5	Compliance with Purchase Order, Specifications, and Drawings	V				
6	Radiographs (if applicable)			D		
7	Prior to Hydrotest	V				
8	List of Lubricants			D		
9	Hydrostatic Test	W				
10	Coating Report and Inspection		R			
11	Shop Visits / Assembly Process Inspections	W				COMPANY inspector to perform shop visits at various stages of the fabrication/assembly process
12	Functional Test				H	
13	Nameplate / Tagging			D		
14	Preparation For Shipment/ Release to Ship				H	
15	As-Built Drawings			D		
16	Auxiliary Equipment Testing					
17	Final Weighing			D		

9. SPECIFICATION

- 22MH194-EM2024 - Performance Specification Water Purification System – Rev. B



PERFORMANCE SPECIFICATION



Client: Electric Hydrogen	Rev A	Date 5/22/23	Issued For Client Review	By MHC	Chk'd PJA	App'd RP	Originated By: M. Heston-Cook	Date: 5/17/23
Project No.: 22MH194	B	5/24/23	Bid	MHC	PJA	RP	Document No.: 22MH194-EM2024	
Project: Product V1 Hydrogen Plant							Page: Page 1 of 17	

**ELECTRIC HYDROGEN
PRODUCT V1 HYDROGEN PLANT
PERFORMANCE SPECIFICATION
WATER PURIFICATION SYSTEM
MAY 24, 2023**

TABLE OF CONTENTS

1. REVISIONS SUMMARY	3
2. INTRODUCTION.....	4
3. SCOPE OF SUPPLY	4
4. GENERAL REQUIREMENTS.....	5
5. PROCESS DESIGN CRITERIA.....	6
6. MECHANICAL DESIGN CRITERIA	7
7. ELECTRICAL & INSTRUMENTATION DESIGN CRITERIA	8
8. COATING REQUIREMENTS.....	10
9. INSPECTION AND TESTING.....	11
10. SHIPMENT.....	11
11. SUPPLIER DATA REQUIREMENTS	12
12. WARRANTY	12
APPENDIX A - WATER QUALITY / SPECIFICATIONS AT SYSTEM INLET.....	13
APPENDIX B - WATER QUALITY / SPECIFICATIONS REQUIRED AT SYSTEM OUTLET.....	14
APPENDIX C - PROCESS FLOW DIAGRAM.....	15
APPENDIX D - PLANT POWER VS RODI CYCLE LENGTH.....	16
APPENDIX E - APPROVED MANUFACTURERS LIST	17

[illegible]

2. INTRODUCTION

M&H Consulting (MHC) has been contracted by Electric Hydrogen to engineer and design their V1 Hydrogen Plant. The V1 Hydrogen Plant uses cell stack(s) to split water into Hydrogen and Oxygen using the electrolysis process. The water used in the electrolysis system requires treatment in a water purification system prior to being introduced into the electrolysis system. MHC is requesting a proposal to provide the water purification system for the plant. This specification outlines the minimum design and performance requirements for the water purification system.

3. SCOPE OF SUPPLY

3.1. Water Purification System

- 3.1.1. SUPPLIER shall engineer, design, fabricate, test, and furnish a complete, fully functional, containerized water purification system including all water purification and utility / ancillary equipment required for operation in accordance with this specification for use in the Hydrogen Plant.
- 3.1.2. The Hydrogen Plant is designed as a low cost, high volume (deployment of multiple facilities on a yearly basis to various customers) facility incorporating a standard design across all of the plants. The water purification system will be integrated into the design of the Hydrogen Plant and therefore must also incorporate a standard design for ease of manufacturing and volume ordering.
- 3.1.3. SUPPLIER shall recommend their preferred pre-treatment process and provide options for pre-treatment if different pre-treatment processes can be recommended for use in the system.
- 3.1.4. SUPPLIER shall incorporate the use of the double pass reverse osmosis (RO) and electrodeionization (EDI) processes in the design of the water purification system. If required SUPPLIER shall incorporate additional water treatment processes into the system design to meet the required water outlet quality, SUPPLIER shall specify and provide details of the additional processes in their proposal.
- 3.1.5. SUPPLIER shall design the system to meet the required outlet water quality using the minimum amount of chemical treatment possible.
- 3.1.6. SUPPLIER shall be responsible for the engineering, design, drafting, fabrication, coating, testing, and documentation of all equipment. These responsibilities include, but are not limited to:
 - All process, mechanical, structural, electrical, instrumentation, and piping engineering and design work to ensure the technical compatibility of all items of equipment within SUPPLIER scope of supply, and to ensure that all guaranteed performance criteria are met

- Development of all required mechanical, structural, electrical, instrumentation, and piping design and fabrication drawings
- Specification and purchasing of all required mechanical, structural, electrical, instrumentation, and piping components. SUPPLIER to utilize only those manufacturers listed in the Approved Manufacturers List (AML), included in Appendix E. Deviations from the AML must be pre-approved by COMPANY
- All necessary project management work, coordinate the activities of all sub-contractors supplying equipment and services, to ensure that all goods and materials are supplied to the agreed project timescale, and in accordance with the required specifications, codes, and standards
- Fabrication, assembly, inspection, hydro and/or pneumatic testing, and painting
- Document and data supply per specifications
- All NDE
- Startup and Commissioning Spares

4. GENERAL REQUIREMENTS

4.1. Design Codes

The following codes shall be interpreted as the minimum requirements governing the subject work (as applicable), and no statement contained in this specification shall be construed as limiting the work to such minimum requirements. The latest editions of the codes listed shall govern all work.

- 4.1.1. ASCE 7, "Minimum Design Loads and Associated Criteria for Buildings and Other Structures"
- 4.1.2. ASME Section VIII, Division 1, "Boiler and Pressure Vessel Code"
- 4.1.3. ASME Section IX, "Welding and Brazing Qualifications"
- 4.1.4. ASME B16.5, "Pipe Flanges and Flanged Fittings"
- 4.1.5. ASME B31.3, "Process Piping"
- 4.1.6. ASME B73.1, "Specification for Horizontal End Suction Centrifugal Pumps for Chemical Process"
- 4.1.7. ASTM, all applicable specifications
- 4.1.8. AWWA, all applicable specifications
- 4.1.9. ISA-5.1-2022, Instrumentation Symbols and Identification
- 4.1.10. NEMA VE2, "Cable Tray Installation Guidelines"

- 4.1.11. NFPA 70, “National Electrical Code”
- 4.1.12. NFPA 70E, “Standard for Electrical Safety in the Workplace”
- 4.1.13. NFPA 101, “Life Safety Code”
- 4.1.14. NSF, all applicable specifications
- 4.1.15. OSHA requirements as applicable
- 4.2. Site Conditions

The water purification system shall be designed in accordance with the following site conditions:

 - 4.2.1. Location: Various locations in the United States
 - 4.2.2. Ambient temperature: -20 to 40 Deg. C
 - 4.2.3. Elevation: Up to 1000 meters
 - 4.2.4. Seismic Design: ASCE 7 Design Category D with SDS >1
 - 4.2.5. Wind Design: Basic Wind Speed = 140 MPH at 0 Ft. Elevation
 - 4.2.6. Ice Design: Maximum of 2.0”

5. PROCESS DESIGN CRITERIA

The following process design criteria shall be used by SUPPLIER to design the water purification system.

Water Purification System Design Criteria:	
Design Water Flowrate at System Outlet	80 GPM continuous
Water Flowrate at System Inlet	SUPPLIER to advise
Water Pressure at System Inlet	35-80 PSIG (regulated by others)
Water Pressure at System Outlet	SUPPLIER to advise (~30 PSIG is required)
Water Temperature at System Inlet	40-100 Deg. F
Water Temperature at System Outlet	SUPPLIER to advise
Water Quality / Specifications at System Inlet	See Appendix A
Water Quality / Specifications Required at System Outlet	Meets ASTM D1193-99e1 Type 1, 18.2 MΩ-cm resistivity – See Appendix B for ASTM Spec.
Total Organic Carbon (TOC) Content at System Outlet	<5 ppb

- 5.1. For reference, a process flow diagram of the Hydrogen plant is located in Appendix C, the water purification system is outlined and tagged X-100. The water outlet from the water purification system will feed into a 3,500-gallon vessel (V-120) which also receives residual water from the plant.

- 5.2. For reference, a graph showing the water purification system expected cycling rate (on/off cycles per hour) versus the H2 Plant stack / input power (MW) is located in Appendix D
- 5.3. For reference, a graph showing the expected fill time (minutes) and drain time (minutes) of vessel V-120 along with the water purification system expected cycle length (minutes) versus the H2 plant stack / input power (MW) is also located in Appendix D.
- 5.4. SUPPLIER shall provide a process flow or block flow diagram of their proposed water purification system with their proposal.

6. MECHANICAL DESIGN CRITERIA

- 6.1. Containerization
 - 6.1.1. The water purification system including all equipment, piping, instrumentation, and electrical power distribution / overcurrent protection equipment shall be mounted in and fully enclosed in ISO shipping container(s).
 - 6.1.2. An area of 22'-6" x 40' has been allotted in the plant layout for the water purification system. This will allow for the installation of two standard 40' long ISO shipping containers.
 - 6.1.3. Containers shall include ISO corner castings for four-point lifting.
 - 6.1.4. Container floors shall be covered with a minimum of ¼" checkered deck plate and seal welded at all edges.
 - 6.1.5. At a minimum, containers shall be provided with one single man door (with panic bar) and one double door.
 - 6.1.6. Container walls shall be insulated.
 - 6.1.7. Containers shall be provided with wall or roof mounted exhaust / ventilation louver(s) with thermostatic control.
 - 6.1.8. Containers shall be provided with wall or roof mounted exhaust fan(s) with thermostatic control to remove heat from the enclosure.
 - 6.1.9. Containers shall be provided with radiant heater(s) with fan(s) and thermostatic control to maintain the temperature inside of the enclosure above 32°F.

6.2. Piping

- 6.2.1. SUPPLIER shall select all piping materials (i.e., thermoplastic, carbon steel, stainless steel, etc.) based on their experience to ensure the water quality requirements are met and shall provide a description of all piping materials in their quote. Piping and piping materials shall be inert to the action of the water they transport and shall preserve the water quality achieved by the system.
- 6.2.2. All process and utility piping shall contain a sufficient number of fittings to allow removal of the piping and equipment for maintenance.
- 6.2.3. Adequate clearance shall be maintained around control valves and in-line instrumentation to permit removal of equipment without additional disassembly of piping. All piping shall be routed and supported for convenience of operational access inside the enclosure, and to provide necessary maintenance clearances.
- 6.2.4. Piping connections shall be provided where needed for breakout of individual equipment or valves.

7. ELECTRICAL & INSTRUMENTATION DESIGN CRITERIA

- 7.1. Available facility electrical supply power is 480V, 3Ph, 60HZ.
- 7.2. COMPANY will provide SUPPLIER with a single point 480V, 3Ph, 60HZ connection to the plant electrical power system per container. SUPPLIER shall provide all electrical power distribution equipment, motor starters, and overcurrent protection equipment required for operation of the system including ancillary 120VAC and 24VDC loads. Each container shall be provided with an incoming circuit breaker.
- 7.3. Equipment and installations shall be suitable for an unclassified (non-hazardous) location. The installations shall adhere to NFPA 70 requirements.
- 7.4. All SUPPLIER installed components and equipment shall be LISTED or labeled by an OSHA recognized NRTL such as UL, FM, or CSAUS. This listing requirement includes custom assemblies, which shall be assembled and labeled by a fabrication facility certified to UL 508. For equipment not bearing the appropriate label, SUPPLIER shall be responsible for the costs and labor required to obtain special approval and shall submit confirming documentation. Equipment that does not comply with this requirement will be rejected.
- 7.5. Containers shall be provided with interior LED lighting including two-way switches for interior lighting located at each door. Minimum interior lighting shall be 20 footcandles at the floor. Emergency lighting, with a minimum 90 minute battery, shall be installed above each doorway. Photocell controlled exterior lighting shall be provided above each doorway.

- 7.6. Container interiors shall be provided with at least one duplex 120V electrical wall outlet per wall.
- 7.7. SUPPLIER shall provide a complete and operational equipment package including instrumentation, junction boxes, control panels, switches, hardware, and wiring.
- 7.8. SUPPLIER shall provide a main PLC control panel with a panel front mounted HMI for operation of the complete system in one of the ISO containers. All instrumentation located in this container shall be wired into the PLC.
- 7.9. The second ISO container shall be supplied with a Remote I/O (RIO) panel which shall communicate with the main PLC in the first container. Instrumentation located in this container shall be wired into the RIO panel.
- 7.10. Variable frequency drives (VFDs) for motors located in the containers shall be SUPPLIER installed and completely pre-wired. Each VFD shall be provided with a line reactor regardless of horsepower rating.
- 7.11. The main control panel shall communicate status and alarms with the plant via Ethernet or Modbus TCP/IP. SUPPLIER shall provide an address table for COMPANY automation contractor to use for integration with the plant PLC.
- 7.12. Instrumentation shall include but not be limited to condition monitoring of the following:
 - Water outlet resistivity
 - Water outlet TOC level
 - Water outlet TDS level
 - Parameters related to system function/diagnostics (e.g., flow, pressure, etc.)
- 7.13. Instrumentation shall provide both visual indication of the process variable as well as automatic alarm and shutdown as required.
- 7.14. Containers shall be provided with a single point instrument air hookup for connection to the plant instrument air system.
- 7.15. Electrical equipment including control stations, relay cabinets, terminal boxes, and motors shall be provided with nameplates.
- 7.16. Motors shall be:
 - TEFC
 - NEMA Design B
 - Class F Insulation
 - Class B Temperature Rise
 - 1.15 Service Factor

- Inverter Duty Rated
 - For motors 25HP and greater, SUPPLIER shall include a 120V/1PH/60hz space heater
- 7.17. Each container shall be provided with two grounding lugs located on opposite corners. Electrical equipment and components shall be bonded to the container structural steel members per NFPA 70, Article 250.
- 7.18. Conduit shall be rigid galvanized steel or EMT for interior installations. Minimum conduit size shall be $\frac{3}{4}$ ".
- 7.19. Transmitters shall be 4-20 mA HART.
- 7.20. Instrumentation tubing and tubing fittings shall be 316L stainless steel and minimum size shall be $\frac{1}{4}$ " with 0.035" wall thickness.

8. COATING REQUIREMENTS

- 8.1. The enclosure and all piping, structural steel, pipe supports, and equipment fabricated and provided by the SUPPLIER shall be coated in accordance with the SUPPLIER's standard coating system. Stainless steel and plastic equipment/piping does not need to be painted.
- 8.2. Equipment that is purchased by SUPPLIER (such as pumps, motors, control valves, etc.) shall be coated with the manufacturers standard coating system.
- 8.3. Final topcoat color of the enclosure shall be "EH2 Green" (#2FA049).
- 8.4. Carbon steel pipe spools, structures, pipe supports, valves, controls, and other equipment items to be coated by the SUPPLIER shall be blasted and painted prior to assembly.
- 8.5. All carbon steel flange surfaces and bolt holes, except for the flange facing, shall be coated prior to assembly.
- 8.6. Pipe, fittings, and structural steel shall not be primer coated until all shop fabrication has been completed.
- 8.7. Final paint coating shall be performed after the equipment has been fully assembled and hydro and / or pneumatically tested.
- 8.8. Instrumentation tubing shall not be installed until after the final coat of paint is applied.
- 8.9. SUPPLIER to advise if any special coatings are required.

9. INSPECTION AND TESTING

9.1. Inspection

- 9.1.1. COMPANY will inspect the material and workmanship of all aspects of the job. All work not found in conformance with standard fabrication practices shall be rejected and replaced or redone at the SUPPLIER's sole expense to the satisfaction of COMPANY.
- 9.1.2. SUPPLIER is responsible for all inspections and NDE testing. The COMPANY reserves the right to inspect all work and witness all tests.
- 9.1.3. SUPPLIER shall complete all NDE testing required for compliance with ASME code and shall provide NDE testing requirements with the Bid.
- 9.1.4. The SUPPLIER shall furnish COMPANY'S inspector with all specified certification on materials and with shop test data verifying that the specifications are being met.
- 9.1.5. The SUPPLIER shall specify all standard shop tests with the Bid. SUPPLIER to submit their standard Inspection and Testing Plan for approval.
- 9.1.6. All pressure vessels shall be hydrotested in accordance with ASME code and pressure and temperature readings taken during the hydrotesting shall be provided as part of the final data package provided by SUPPLIER.
- 9.1.7. All pressure vessels to be Code Stamped per ASME Sec VIII, Div. 1 and NBR registered.

10. SHIPMENT

10.1. General

- 10.1.1. After all testing is complete, and approval by COMPANY, SUPPLIER shall prepare the system for shipment.

10.2. Preparation for Shipment

- 10.2.1. All openings, flange faces, or screwed connections left open for shipment, shall be coated with a rust inhibitor, and protected with suitable wooden flange covers or plastic caps.

11. SUPPLIER DATA REQUIREMENTS

11.1. Supplier Data Requirements

11.1.1. SUPPLIER shall be responsible for supplying data in accordance with the Supplier Data Requirements List (SDRL).

11.2. Material Traceability

11.2.1. SUPPLIER shall keep a record and provide traceability of all equipment, piping, instrumentation, electrical, and structural materials.

12. WARRANTY

SUPPLIER shall guarantee all work to be free from defect in design, workmanship, materials, and equipment. SUPPLIER shall repair or reinstall, without cost to COMPANY, any work, material, or equipment furnished by him that shall become defective, or not operate in accordance with the design requirements except for wear and tear, during the progress of the work or during a period of 12 months after successful start-up, or 18 months after acceptance by COMPANY.

The SUPPLIER shall guarantee the performance of the system per the design conditions specified for the duration of the warranty period.

Under this warranty, the SUPPLIER shall replace, at his expense, all materials and/or parts that are found to be defective or which do not meet the specified design conditions during the guarantee period. The SUPPLIER shall also furnish (at his expense) an experienced service representative to supervise the necessary repairs and replacements.

APPENDIX A - WATER QUALITY / SPECIFICATIONS AT SYSTEM INLET

For product available in 2024

100MW Electrolyzer Plant

Input Water Standard

Included within EH2's product is a Reverse Osmosis/Electro-Deionization unit to ensure sufficient process water purity. The table below lists limits to influent water parameters such that required purity will be reached without the need for additional treatment equipment.

Water Quality Limits

Parameter	Specification	Units
Alkalinity (as CaCO ₃)	< 300	mg/L
Aluminum	< 1.6	mg/L
Arsenic	< 0.01	mg/L
Barium	< 2	mg/L
Beryllium	< 0.004	mg/L
Bicarbonate	< 150	mg/L
Boron	< 1	mg/L
Cadmium	< 0.005	mg/L
Calcium	< 100	mg/L
Carbon Dioxide, Free/Residual	< 7	mg/L
Chloride	< 250	mg/L
Chlorine, Free/Residual	< 5	mg/L
Chromium	< 0.1	mg/L
Copper	< 1.3	mg/L
Fluoride	< 4	mg/L
Hardness (as CaCO ₃)	< 435	mg/L
Iron	< 1.5	mg/L
Lead	< 0.015	mg/L
Lithium	< 0.01	mg/L
Magnesium	< 45	mg/L
Manganese	< 0.3	mg/L
Nickel	< 0.1	mg/L
Nitrate	< 10	mg/L
Nitrite	< 1	mg/L
pH	6.5 - 9.7	-
Potassium	< 8.5	mg/L
Selenium	< 0.05	mg/L
Silica	< 20	mg/L
Silver	< 0.1	mg/L
Sodium	< 175	mg/L
Strontium	< 0.225	mg/L
Sulfate	< 250	mg/L
Total Dissolved Solids (TDS)	< 500	mg/L
Total Organic Carbon (TOC)	< 8	mg/L
Turbidity	< 7	NTU
Vanadium	< 0.05	mg/L

APPENDIX B - WATER QUALITY / SPECIFICATIONS REQUIRED AT SYSTEM OUTLET



Standard Specification for Reagent Water¹

This standard is issued under the fixed designation D 1193; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ε) indicates an editorial change since the last revision or reapproval.

This standard has been approved for use by agencies of the Department of Defense.

^{ε1} NOTE—An editorial change was made in 1.1 in October 2001.

1. Scope

1.1 This specification covers requirements for water suitable for use in methods of chemical analysis and physical testing. Four grades are specified:

	Type I	Type II	Type III	Type IV
Electrical conductivity, max, $\mu\text{S}/\text{cm}$ at 298 K (25°C)	0.056	1.0	0.25	5.0
Electrical resistivity, min, $\text{M}\Omega\cdot\text{cm}$ at 298 K (25°C)	18	1.0	4.0	0.2
pH at 298 K (25°C)	^A	^A	^A	5.0 to 8.0
Total organic carbon (TOC), max, $\mu\text{g}/\text{L}$	50	50	200	no limit
Sodium, max, $\mu\text{g}/\text{L}$	1	5	10	50
Chlorides, max, $\mu\text{g}/\text{L}$	1	5	10	50
Total silica, max, $\mu\text{g}/\text{L}$	3	3	500	no limit
Microbiological contamination—When bacterial levels need to be controlled, reagent grade types should be further classified as follows:				
		Type A	Type B	Type C
Maximum heterotrophic bacteria count		10/1000 mL	10/100 mL	100/10 mL
Endotoxin, EU/mL ^B		<0.03	0.25	not applicable

^A The measurement of pH in Type I, II, and III reagent waters has been eliminated from this specification because these grades of water do not contain constituents in sufficient quantity to significantly alter the pH.

^B EU = Endotoxin Units.

1.2 The method of preparation of the various grades of reagent water determines the limits of impurities and shall be as follows:

1.2.1 Type I grade of reagent water shall be prepared by distillation or other equal process, followed by polishing with a mixed bed of ion exchange materials and a 0.2- μm membrane filter. Feedwater to the final polishing step must have a maximum conductivity of 20 $\mu\text{S}/\text{cm}$ at 298K (25°C).

1.2.2 Type II grade of reagent water shall be prepared by distillation using a still designed to produce a distillate having a conductivity of less than 1.0 $\mu\text{S}/\text{cm}$ at 298 K (25°C). Ion exchange, distillation, or reverse osmosis and organic adsorption may be required prior to distillation if the purity cannot be attained by single distillation.

NOTE 1—Because distillation is a process commonly relied upon to produce high purity water, the levels specified for Type II reagent water were selected to represent the minimum quality of water that a distillation process should produce.

1.2.3 Type III grade of reagent water shall be prepared by distillation, ion exchange, continuous electrodeionization reverse osmosis, or a combination thereof, followed by polishing with a 0.45- μm membrane filter.

1.2.4 Type IV grade of reagent water may be prepared by distillation, ion exchange, continuous electrodeionization reverse osmosis, electrodialysis, or a combination thereof.

1.3 The choice of one of the various grades may be designated by the method or by the investigator.

1.4 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.*

2. Referenced Documents

2.1 ASTM Standards:

D 1125 Test Methods for Electrical Conductivity and Resistivity of Water²

D 1129 Terminology Relating to Water²

D 1293 Test Methods for pH of Water²

D 4453 Practice for Handling of Ultra-Pure Water Samples²

D 4517 Test Method for Low-Level Total Silica in High-Purity Water by Flameless Atomic Absorption Spectroscopy^{3, 4}

D 4779 Test Method for Total, Organic, and Inorganic Carbon in High Purity Water by Ultraviolet (UV) or Persulfate Oxidation, or Both, and Infrared Detection⁴

D 5391 Test Method for Electrical Conductivity and Resistivity of a Flowing High Purity Water Sample²

D 5542 Test Method for Trace Anions in High Purity Water by Ion Chromatography²

¹ This specification is under the jurisdiction of ASTM Committee D-19 on Water and is the responsibility of Subcommittee D19.02 on General Specifications, Technical Resources, and Statistical Methods.

Current edition approved Feb. 10, 1999. Published March 1999. Originally issued as D 1193 – 51 T. Last previous edition D 1193 – 91.

² *Annual Book of ASTM Standards*, Vol 11.01.

³ Determination of Trace Silica in Industrial Process Waters by Flameless Atomic Absorption Spectrometry, Judith Rawa and Earl Henn, *Analytical Chemistry*, Vol 51, No 3, March 1979.

⁴ *Annual Book of ASTM Standards*, Vol 11.02.

D 5997 Test Method for On-Line Monitoring of Total Carbon, Inorganic Carbon in Water by Ultraviolet, Persulfate Oxidation and Membrane Conductivity Detection²

D 6071 Test Method for Low Level Sodium in High Purity Water by Graphite Furnace Atom Absorption Spectroscopy²

F 1094 Test Methods for Microbiological Monitoring of Water Used for Processing Electron and Microelectronic Devices by Direct Pressure Tap Sampling Valve and by the Presterilized Plastic Bag Method⁵

3. Terminology

3.1 *Definitions*— For definitions used in this specification refer to Terminology D 1129.

4. Significance and Use

4.1 Different analytical methods and industrial processes require water of different purities. Also, different types of contaminants affect these methods and processes differently. This specification is intended to provide the user with a choice of different grades of water having different purities as described in Section 1. It is intended to satisfy the requirements for normal laboratory procedures, but does not necessarily apply to the large-scale production of pure water or for specific applications. Different grades of water may be covered by other ASTM specifications or guides, or by the requirements of other standards organizations.

5. Use and Application

5.1 The method of preparing Type I reagent water may add organic contaminants to the water by contact with the ion-exchange materials. It should be noted also that the method may or may not remove non-ionized dissolved gases.

5.1.1 The dissolved or particulate organic contamination would normally range from 30 to 50 µg/L. The concentration of nonionized dissolved gases may exceed 5 mg/L.

5.2 The quality of the effluent water depends upon the type, age, and method of regeneration of the ion exchange materials. Likewise, the flow rate through the ion exchange resin bed will change the conductivity of the effluent water. The instructions of the manufacturer of the resins or the resin cartridge bed should be followed.

5.3 The use of the membrane filter in the preparation of Type I and Type III water may add a small amount of organic components to the water first produced.

5.3.1 Some membrane filters contain as much as 8 mass % of soluble components resulting from the manufacturing process.

5.3.2 If the contamination of the water by the organic component is of significance to the test, Type II water should be used or the membrane should be rinsed by discarding the first 10 mL of water produced per square centimetre of filter area, and until a test for the organic components shows them to have been reduced to less than the specified level.

5.4 Type II grade of reagent water should be sterile and pyrogen-free as produced and generally may be used whenever

freedom from organic or biological contaminants is desirable. However, the method of storage and handling of the water may itself result in contamination.

5.4.1 Type II water should be pyrogen-free, but must be tested in conformance with the requirements of the current edition of U.S.P. if proof is needed.

5.4.2 The description of Type II reagent water is intended to characterize the product of distillation.

5.4.3 Distilled water is often specified when freedom from moles and silica is required, or when surface-active organics are proven to be a problem, and when alternative processes have been found to be inadequate.

5.5 Types I, II, and III reagent water should be protected from atmospheric contamination and from solution of container and tubing materials.

5.5.1 Extreme care must be exercised in handling samples when making an analysis. Sample containers and tubing should be made of fluorocarbon, titanium, tantalum, block tin, quartz, 18-8 stainless steel, polyethylene, or other material proven to be sufficiently resistant to chemical attack so as not to cause contamination in the intended use. Practice D 4453 should be consulted.

5.6 Because atmospheric gases and impurities rapidly recontaminate exposed water, in-line electrodes should be employed for determining the electrical conductivity of reagent water Types I, II, and III. The measurement of pH in Type I, II, and III reagent waters has been eliminated from this specification because the values would be a function of the instrumentation.

5.7 Since freedom from biological contaminants may be important in the test procedure using any of the reagent waters specified, a classification of bacterial levels is included and should be specified if it is of significance to the test being performed.

5.7.1 To obtain sterile water, any of the types of reagent water listed in this section may be produced, bottled, and heated to 394 K (121°C) for 20 min. This procedure is most easily carried out by autoclaving at 103 kPa (15 psi) for 20 min.

5.8 *The following requirement is beyond the requirements of the general specifications:* The use of reagent grade water should recognize that analytes may exist in water that meets the criteria listed in Section 1, but these analytes may interfere with the use of the water. If levels of other analytes are important, it is the user's responsibility to specify their limits.

6. Requirements

6.1 Reagent water shall conform to the requirements specified in Section 1.

6.2 Additional requirements concerning specific contaminants or methods of preparation may be included in this specification by mutual agreement between the parties concerned.

7. Test Methods

7.1 *Electrical Conductivity and Resistivity*—Refer to Test Methods D 1125 and D 5391.

7.2 *pH*—Refer to Test Methods D 1293.

7.3 *Silica*—Refer to Test Method D 4517.

⁵ Annual Book of ASTM Standards, Vol 10.04.

7.4 *Sodium*—Refer to Test Methods D 6071.
7.5 *Chlorides*⁶. Refer to Test Method D 5542.
7.6 *TOC*—Refer to Test Method D 4779 and D 5997.

7.7 *Endotoxins*—Refer to LAL Test Method.⁶
7.8 *Microbiological Contamination*—Refer to Test Methods F 1094.

8. Keywords

8.1 laboratory analysis; reagent; water

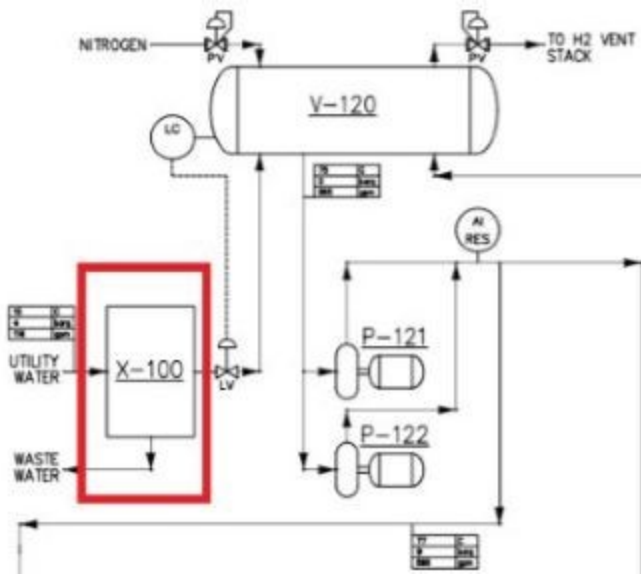
⁶ Published in *U.S. Pharmacopeia* by The U.S. Pharmacopeial Convention, Inc.

ASTM International takes no position respecting the validity of any patent rights asserted in connection with any item mentioned in this standard. Users of this standard are expressly advised that determination of the validity of any such patent rights, and the risk of infringement of such rights, are entirely their own responsibility.

This standard is subject to revision at any time by the responsible technical committee and must be reviewed every five years and if not revised, either reapproved or withdrawn. Your comments are invited either for revision of this standard or for additional standards and should be addressed to ASTM International Headquarters. Your comments will receive careful consideration at a meeting of the responsible technical committee, which you may attend. If you feel that your comments have not received a fair hearing you should make your views known to the ASTM Committee on Standards, at the address shown below.

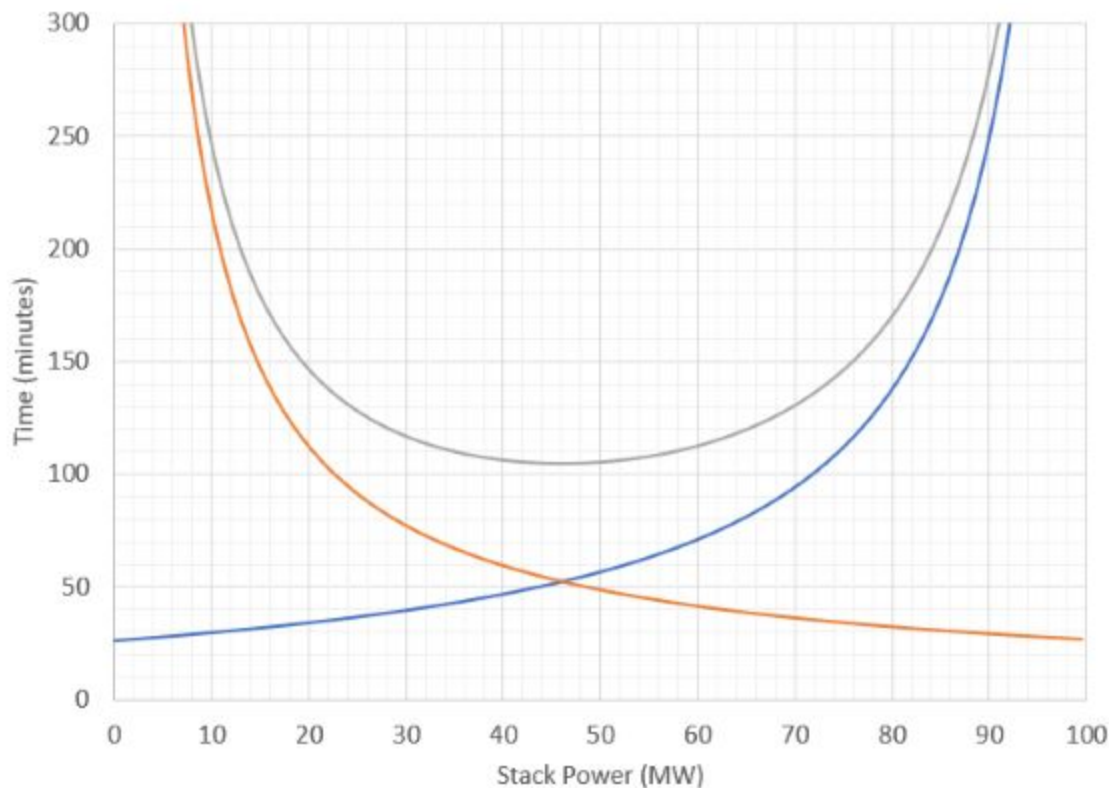
This standard is copyrighted by ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA 19428-2959, United States. Individual reprints (single or multiple copies) of this standard may be obtained by contacting ASTM at the above address or at 610-832-9585 (phone), 610-832-9555 (fax), or service@astm.org (e-mail); or through the ASTM website (www.astm.org).

APPENDIX C - PROCESS FLOW DIAGRAM



APPENDIX D - PLANT POWER VS RODI CYCLE LENGTH

RO/DI Tank Fill Time, Drain Time, and Cycle Length vs. Stack Power

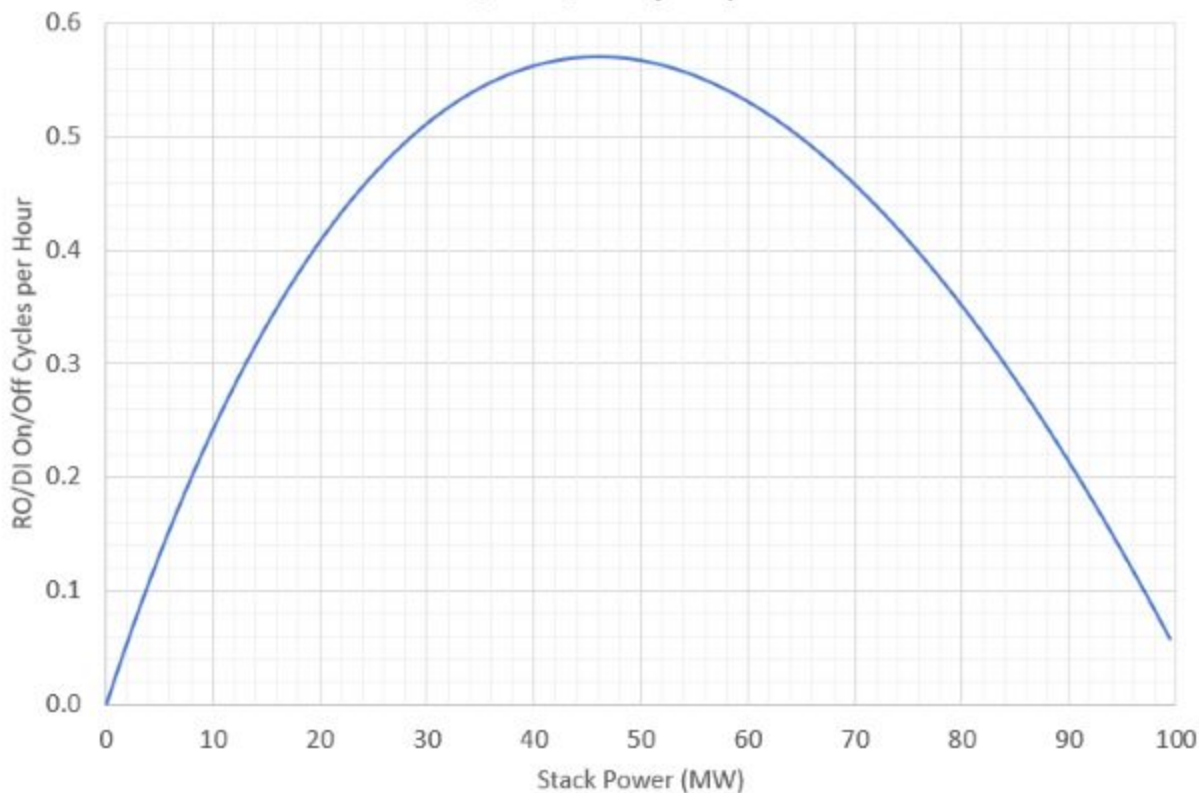


— Tank filling (RO/DI unit 'On')
 — Tank draining (RO/DI unit 'Off')
 — RO/DI Cycle Length (On->Off)

Inputs:

- Tank Volume = 3500 gal
- 'Filled' at 80% water level
- 'Empty' at 20% water level

RO/DI On/Off Cycles per Hour vs. Stack Power



Inputs:

- Tank Volume = 3500 gal
- 'Filled' at 80% water level
- 'Empty' at 20% water level
- RO/DI Output = 80 GPM

APPENDIX E - APPROVED MANUFACTURERS LIST

<u>Flow</u>		
CH/XC/XCV	Chokes – Fixed/Adjustable	Varco/Best
FE	Flow Element, Senior Orifice	Daniel
FE	Flow Element, Flange/Orifice Union	Daniel
FE/RO	Flow Element, Fixed/Restriction Orifice	Daniel
FE	Flow Element, V-Cone	Nuflo
FE	Flow Element, Ultrasonic	Daniel
FE	Flow Element, Ultrasonic (Flare)	Sick
FE	Flow Element, Rotometer (Variable Area)	Brooks
FE	Flow Element, Turbine	Nuflo/Barton
FE	Flow Element, PD	FMC Smith Meter
FE	Flow Element, Coriolis	Micro Motion
FE	Flow Element, Vortex	Rosemount
FE	Flow Element, Mag	Rosemount
FE	Flow Element, Thermal	FCI
FI	Sight Flow Indicator	Brooks
FIC	Flow Indicating Controller, Pneumatic	Fisher
FR	Flow Recorder, Chart	Barton
FV/FCV	Flow Control Valve	Fisher
FSH/FSL	Flow Switch	FCI
FT	Flow Transmitter, dP	Rosemount
FT	Flow Transmitter, Pulse	Nuflo, FMC Smith Meter
FT/UT	Flow Transmitter, Multivariable	Rosemount
FY	I/P Transducer, Smart	Fisher
<u>Level</u>		
LC	Level Controller, Pneumatic	Fisher
LC/LT	Level Controller, Electric (Float)	Fisher
LC/LT	Level Controller, Electric (dP)	Rosemount
LC/LT	Level Controller, Electric (Guided Wave)	Rosemount
LC/LT	Level Controller, Electric (Non-Contact)	Rosemount
LC/LT	Level Controller, Electric (Mag)	Magnetrol
LG	Level Gauge, Magnetic	Magnetrol
LG	Level Gauge, Armored	Quest-Tec
LG	Tank Gauges	Rosemount Tank Gauging
LSH/LSL	Level Switch, Electric (Float)	Fisher
LSH/LSL	Level Switch, Electric (Vibrating Fork)	Rosemount

LSH/LSL	Level Switch, Pneumatic	Fisher
LV/LCV	Level Control Valve	Fisher
LY	I/P Transducer, Smart	Fisher
<u>Pressure</u>		
PC	Pressure Controller, Pneumatic	Fisher
PCV	Pressure Regulator	Fisher
PCV	Pressure Regulator, Low Pressure	Fisher
PCV	Filter Regulator	Fisher
PDI	Differential Pressure Gauge	Orange Research
PDT	Differential Pressure Transmitter	Rosemount
PDC	Differential Pressure Controller	Fisher
PI	Pressure Gauge	Ashcroft
PR	Pressure Recorder, Chart	Barton
PSE	Rupture Disc	Fike
PSH/PSL	Pressure Pilot, Pneumatic Panel Mount	Versa
PSH/PSL	Pressure Switch, Electric	ITT
PSV	Pressure Safety Valve	Anderson-Greenwood/Crosby
PSV	Pressure Safety Valve, Panel	Nupro
PSV	Pressure Safety Valve, Low Pressure	Enardo
PVSV	Pressure Vacuum Safety Valve	Enardo
PV/PCV	Pressure Control Valve	Fisher
PT	Pressure Transmitter	Rosemount
PY	I/P Transducer, Smart	Fisher
<u>Temperature</u>		
TC	Temperature Controller, Pneumatic	Fisher
TE	Temperature Element, RTD	Rosemount
TE	Temperature Element, Thermocouple	Rosemount
TI	Temperature Gauge	Ashcroft
TR	Temperature Recorder, Chart	Barton
TSH/TSL	Temperature Switch, Electric	ITT
TSH/TSL	Temperature Switch, Pneumatic	EXLINE Exhaust
TT	Temperature Transmitter	Rosemount
TW	Thermowell	Rosemount
TV/TCV	Temperature Control Valve	Fisher
TY	I/P Transducer, Smart	Fisher

<u>Miscellaneous</u>		
AE	Moisture Analyzer	SpectraSensors
AE	Fire Detector	Det-tronics
AE	Gas Detector	Det-tronics
AE	Smoke Detector	Det-tronics
AE	BS&W Monitor	FMC
AE	Oil in Water Analyzer	FMC
AE	Sample Probe	A+ Corp (Genie)
AE	Gas Chromatograph	Daniel
AE	Corrosion Coupon	Corrpro
AE	Turbidity Monitor	Rosemount
BMS	Burner Management System	Profire
DV	Diverter Valve, 3-way	KTM
FA	Flame Arrestor	Enardo
MCC	Motor Control Center	Allen-Bradley
PLC	Programmable Logic Controller	Allen-Bradley
RTU	Remote Terminal Unit (Flow Computer)	ABB Totalflow
SDV	Shutdown Valve	WKM
SDV	Shutdown Valve, Pneumatic Actuator	Bettis
SDV	Shutdown Valve, Electric Actuator	Limitorque
SDY/SY/BDY	Solenoid Valve	Versa
TH	Thief Hatch	Enardo
VFD	Variable Frequency Drive	Allen-Bradley
ZSC/ZSO	Limit Switch	Westlock